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Institute of Health Policy, Management and Evaluation
UNIVERSITY OF TORONTO

Dalla Lana
School of Public Health

2

3

PARAMETRIC-METRIC

$$\text{Life Expectancy} = \beta_0 + \beta_1 \times \text{GDP} + \beta_2 \times \text{Clean Water Access} + \beta_3 \times \text{Education Level} + \beta_4 \times \text{Vaccination Rate} + \epsilon$$

“New Global Dashboard on COVID-19 Vaccine Equity finds low-income countries would add \$38 billion to their GDP forecast for 2021 if they had the same vaccination rate as high-income countries. Global economic recovery at risk if vaccines are not equitably manufactured, scaled up and distributed.”



**World Health
Organization**

Source

PARAMETRIC VS NON-PARAMETRIC MODELS

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PARAMETRIC VS NON-PARAMETRIC MODELS

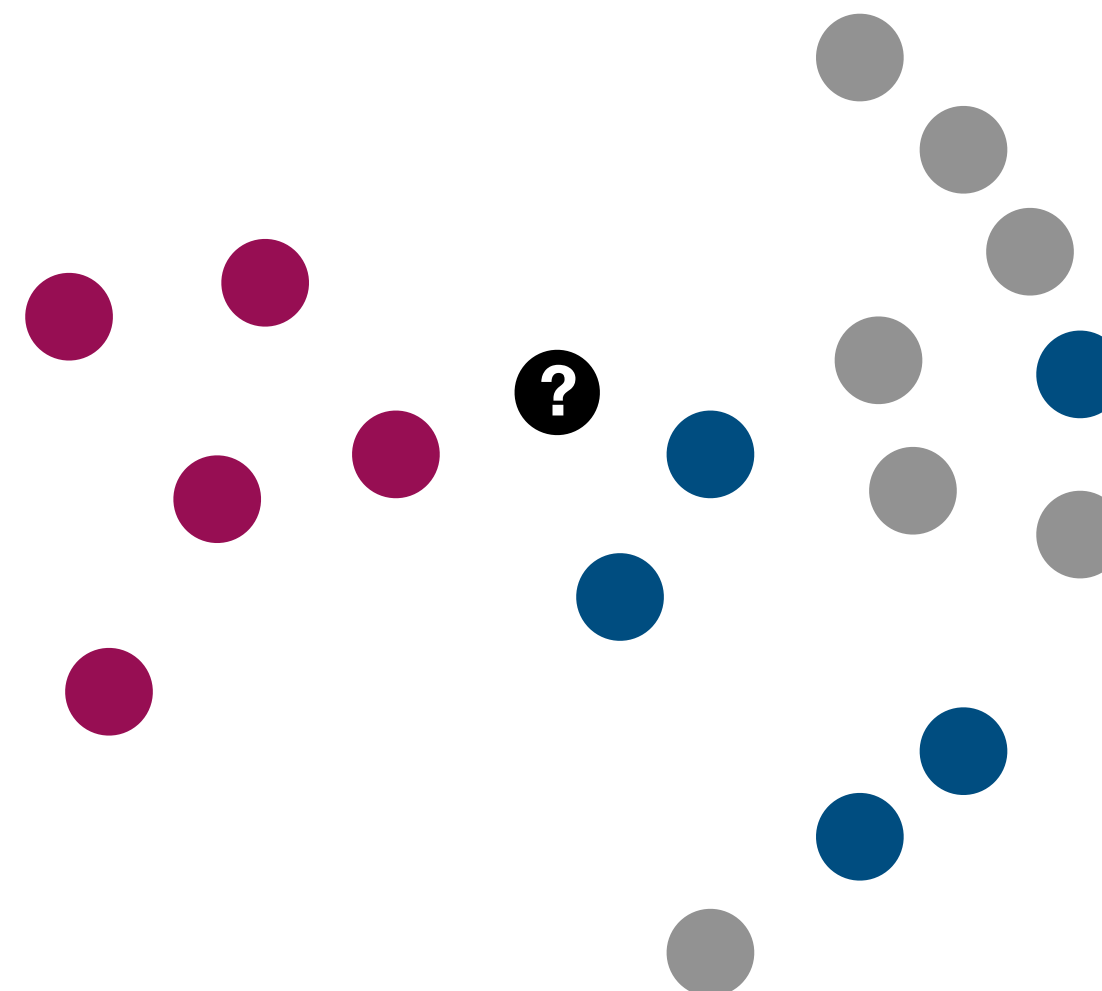
Example

Scenario: Imagine we have a dataset of different countries. For each country, we have information on GDP, education level, access to clean water, and vaccination rates. We also have the average life expectancy for each country.

Goal: For a new country (or a region within a country) with known GDP, education level, access to clean water, and vaccination rates, we want to predict the life expectancy (Continuous or Discrete).



- GDP: \$40,000
- Education Level (Average Years of Schooling): 10 years
- Access to Clean Water: 90%
- Vaccination Rate: 80%



Life Expectancy

- High
- Average
- Low